distance that appears in Newton's gravitational law? Either choice, Mercury's frame or the Sun's, would violate the principle of relativity. This quandary convinced Einstein that Newton's gravitational law must be flawed.

Einstein's audacity is breathtaking. Having discarded Newton's absolute space and absolute time with almost no experimental justification, he was now inclined to discard Newton's enormously successful law of gravity, and with even less experimental justification. However, he was motivated not by experiment, but by his deep, intuitive insight into how the laws of physics *ought* to behave.

Einstein began his search for a new law of gravity in 1907. His initial steps were triggered and guided by a writing project: Although the patent office now classified him as only a "technical expert second class" (recently promoted from third class), he was sufficiently respected by the world's great physicists to be invited to write a review article for the annual publication *Jahrbuch der Radioaktivität und Elektronik* about his special relativistic laws of physics and their consequences. As he worked on his review, Einstein discovered a valuable strategy for scientific research: The necessity to lay out a subject in a self-contained, coherent, pedagogical manner forces one to think about it in new ways. One is driven to examine all the subject's gaps and flaws, and seek cures for them.

Gravity was his subject's biggest gap; special relativity, with its inertial frames on which no gravitational force can act, was totally ignorant of gravity. So while Einstein wrote, he kept looking for ways to incorporate gravity into his relativistic laws. As happens to most people immersed in a puzzle, even when Einstein wasn't thinking directly about this problem, the back of his mind mulled it over. Thus it was that one day in November 1907, in Einstein's own words, "I was sitting in a chair in the patent office at Bern, when all of a sudden a thought occurred to me: 'If a person falls freely, he will not feel his own weight.'"

Now you or I could have had that thought, and it would not have led anywhere. But Einstein was different. He pursued ideas to their ultimate ends; he wrung from them every morsel of insight that he could. And this idea was key; it pointed toward a revolutionary new view of gravity. He later called it "the happiest thought of my life."